

The Hjorth and band power features are then computed from the enhanced EEG signals. These features are then classified into left and right hand MIs using an LDA classifier. A block diagram representation of the proposed method is shown in Fig.1.

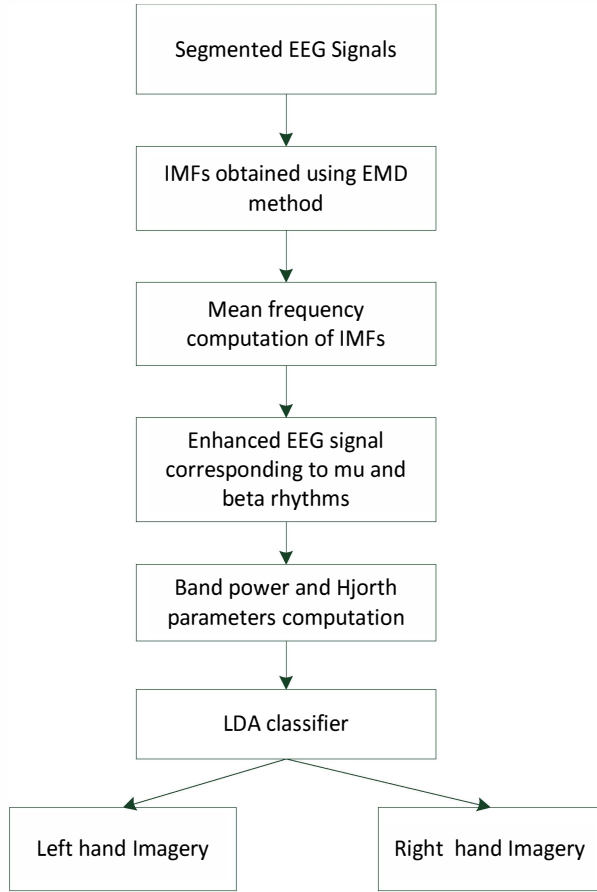


Fig. 1. Block diagram for the proposed method.

II. EMPIRICAL MODE DECOMPOSITION (EMD)

The Empirical mode decomposition (EMD) method decomposes any time-domain signal into a set of AM-FM components without the assumption of stationarity and linearity of the signal. The decomposition based on the EMD is thus an adaptive and signal-dependent decomposition. It is based on the concept that the signal under analysis is a superposition of intrinsic mode functions (IMFs) which are extracted using the sifting process in the EMD method. It should be noted that the conventional signal processing techniques for EEG signal analysis based on the Fourier and wavelet transforms use predefined set of basis functions, which result into poor time-frequency localization [11]. The fixed linear basis functions or *a priori* designed basis functions are useful only for stationary signals and can be sub-optimal for analysis of non-stationary signals like EEG [12], [14]–[17]. In biological systems like a human brain, the frequency of oscillations may not be fixed and drift takes place in EEG rhythms within different frequency bands, and it thus makes the conventional methods like Fourier and wavelet analysis, which use predefined fixed basis functions, inappropriate for EEG signal analysis [15]. For biomedical signals like EEG, it has also been shown that

the better localization of time-varying frequency components of μ and β rhythms during MI is obtained using the EMD method as compared to short-time Fourier transform (STFT) and wavelet transform based methods [14]. The EMD method automatically decomposes a signal $x(t)$ into a finite set of IMFs $D_p(t)$, which can be considered band-limited and symmetric functions [11]. Symmetric nature of IMFs have been explored for classification of epileptic seizure related EEG signals [18]–[20]. Each extracted IMF must satisfy two basic conditions: (i) the number of extrema and the number of zero crossings must differ by at most one or be the same, (ii) at any point, the mean value of the envelopes defined by the local maxima and defined by the local minima is zero. The EMD algorithm [11],[21] for a signal $x(t)$ can be summarized by the following sifting process:

- (1) Assume $g_1(t) = x(t)$.
- (2) Determine the extrema (maxima and minima) of $g_1(t)$.
- (3) Compute the upper and lower envelopes $E_{max}(t)$ and $E_{min}(t)$, respectively, by interpolating the maxima and minima respectively with cubic spline interpolation.
- (4) Compute the local mean as: $m(t) = (E_{max}(t) + E_{min}(t))/2$.
- (5) Subtract $m(t)$ from the original signal as: $g_1(t) = g_1(t) - m(t)$.
- (6) Check whether $g_1(t)$ is an IMF by applying the above mentioned two basic conditions of IMF.
- (7) Repeat steps from (2) to (6) until an IMF $g_1(t)$ is determined.

When the first IMF is determined, then consider $D_1(t) = g_1(t)$, which can be considered the smallest temporal scale in the signal $x(t)$. In order to determine the remaining IMF components, find the residue $rs_1(t)$ of the data by subtracting $D_1(t)$ from the signal as: $rs_1(t) = x(t) - D_1(t)$. The sifting process will be continued until the final residue becomes a function from which the extraction of an IMF is not possible. In the entire sifting process, the basis functions and the residues can be expressed as:

$$rs_1(t) - D_2(t) = rs_2(t), \dots, rs_{M-1}(t) - D_M(t) = rs_M(t), \quad (1)$$

where $rs_M(t)$ is the final residue. In the end of the entire sifting process, the signal $x(t)$ can be expressed as a linear combination of IMFs and a residue as follows [11]:

$$x(t) = \sum_{p=1}^M D_p(t) + rs_M(t), \quad (2)$$

where M is the number of IMFs and $rs_M(t)$ is the final residue. It should be noted that the IMFs are narrow-band components of the signal. In order to obtain an enhanced EEG signal corresponding to μ and β rhythms, from the original EEG signal, selection of an optimal number of IMFs is required. In this work, we have proposed the selection of

IMFs based on the mean frequency computation from these IMFs so as to obtain an enhanced EEG signal corresponding to μ band (8-13 Hz) and β band (14-24 Hz). The enhanced EEG signals are obtained from the summing of these IMFs whose mean frequencies belong to μ and β bands. In order to cover both the rhythms (μ and β), we have considered (6-26 Hz) frequency range for selection of IMFs of EEG signals.

III. FEATURE EXTRACTION

The features namely, Hjorth and band power features are commonly used features for classification of left and right MI EEG signals for BCI applications [1], [22], [23], [24]. In this work, we have considered both of these features extracted from the enhanced EEG signals for classifying MI EEG signals.

The first Hjorth parameter, namely activity is the measure of the average power of the signal (variance of the signal). Mathematically, it can be expressed as [25]:

$$Activity = \sum_{i=1}^{M_s} \frac{(p(i) - \mu)^2}{M_s} \quad (3)$$

The second Hjorth parameter known as mobility, is an estimate of the mean frequency. It can be defined as follows [25]:

$$Mobility = \sqrt{\frac{var(p')}{var(p)}} \quad (4)$$

The third Hjorth parameter known as complexity, is an estimate of the bandwidth of the signal. It can be defined as follows [25]:

$$Complexity = \frac{Mobility(p')}{Mobility(p)} \quad (5)$$

where, p is the signal and p' is the first derivate of the signal, and μ is the mean of the signal in the computation sampling window and M_s is the number of samples in the window. The above mentioned Hjorth features have been computed from a 1 second window of EEG signals from the two channels namely C3 and C4 respectively. We have used all these three Hjorth parameters as input features for an LDA classifier.

We have also considered band power features which are computed over a small time window of 1 second of EEG signals in this work. Typically, the band powers of the two frequency bands namely μ and β are computed for classification of EEG signals corresponding to left and right hand MI tasks. The frequency ranges (8-13 Hz) and (18-25 Hz) have been selected corresponding to μ and β bands respectively [2],[26]. We have selected these bands for computing features as they are more reactive during a cued MI [27] in the form of event related de-synchronization (ERD) and event related synchronization (ERS). The combined features based on Hjorth parameters and band powers have also been used as a final feature set for classification of left and right hand MI EEG signals.

IV. LINEAR DISCRIMINANT ANALYSIS (LDA) CLASSIFIER

Generally, it is not straightforward to classify the extracted features for classification of EEG signals in BCI applications. Finding the optimum combination of the features which can provide better discrimination, is a crucial task. In this paper, we have used an LDA classifier which is commonly used in EEG-based BCI applications. The LDA based classifier tries to reduce the dimensionality and at the same time protect most of the class discriminatory information. Suppose, we have a set of two classes denoted by w_1 and w_2 . Then, we classify the n -dimensional sample points $x = \{x_1, x_2, x_3, \dots, x_n\}$, n_1 samples to the class w_1 , and n_2 samples to the class w_2 . In this method, we try to achieve a line $y = w^t x$ from the set of all possible lines, and that line maximizes the discrimination between the two considered classes. In order to obtain a good projection vector, we require to measure the separation between the two classes chosen for the study. The mean vector of each class in x-space and y-space is given by following equations [28]:

$$\mu_i = \frac{1}{N_i} \sum_{x \in w_i} x, \quad (6)$$

$$\text{and } v_i = \frac{1}{N_i} \sum_{y \in w_i} y = \frac{1}{N_i} \sum_{y \in w_i} w^t x = w^t \mu_i \quad (7)$$

The objective function is defined as the distance between the two projected means. It can be expressed as follows [28]:

$$J(w) = |v_1 - v_2| = |w^t(\mu_1 - \mu_2)| \quad (8)$$

However, the distance measured between projected means may not always be a good measure because the standard deviation between classes has not been considered. In order to overcome the limitation stated above, the enhancement of LDA has been proposed which is known as Fishers LDA classifier. It determines a decision boundary or probably a hyperplane in the feature space to classify the features in to distinct classes. It finds out the separation boundary between two given distributions in terms of the ratio of two group variances as given below [3]:

$$J(w) = \frac{\sigma_{between}^2}{\sigma_{within}^2} = \frac{w^t(\mu_1 - \mu_2)^2}{w^t S_1 w + w^t S_2 w} \quad (9)$$

where μ_1, μ_2 are the mean of the classes and S_1, S_2 are the variances of the feature distributions between two classes w_1, w_2 respectively. The maximum separation between two classes can be shown by (10) as:

$$w^* = (S_1 + S_2)^{-1}(\mu_1 - \mu_2) \quad (10)$$

The w^* is weight vector which provides optimum direction of projection of the data. In the Fisher's LDA, the decision boundary uses the following equation to classify the feature vector $d(m)$ as [29]:

$$p(m) = d(m)w^t + b \quad (11)$$

where b is the threshold or bias. The features are assigned to one of the classes based on the sign of the $p(m)$.

V. RESULTS AND DISCUSSION

In order to evaluate the performance of the proposed method, the BCI competition IV dataset 2B [30] has been used in this work. The dataset contains EEG signals from 9 healthy subjects, denoted by B01-B09. In this dataset 5 sessions has been provided for each subject. The EEG signals recorded from all nine subjects have been used to study in order to show the effectiveness of the proposed method. The data were recorded from three bipolar channels including, C3, Cz, and C4 with a sampling frequency of 250 Hz. In order to evaluate the proposed method, we selected the data from C3 and C4 channels related to the sensorimotor areas. There are different number of trials in each session, e.g., 60, 70 or 80. Each trial involved a paradigm period of 8 second [30]. In this paper, for training phase, a single session namely ^03T has been used. For the evaluation phase, we have used two sessions namely, ^04E and ^05E for computing the accuracy in the classification of left and right MI EEG signals. It should be noted that, the ^ in the session name denotes the subject number which ranges from B01 to B09.

We have also used BCI competition IV dataset 2A [31] for evaluating the performance of the proposed method. This dataset contains EEG recordings from the nine healthy subjects, namely (A01-A09). The EEG signals of all the nine subjects have been used to evaluate the performance of the proposed method. For each of the nine subjects, the data recorded over two sessions are provided, e.g., A01T and A01E [31].

In our study, we have used EEG signals from only two channels namely C3, and C4 respectively. For each of the subjects, in order to compute the classification accuracy (in %), the LDA classifier has been trained with 100% data from the B0 β 03T and tested/evaluated on the 100% data for each of the sessions, B0 β 04E and B0 β 05E, where β is the subject number. Since the MI task starts at 3 second, we have trained and tested the LDA classifier with the features corresponding to EEG signals from 3 second to 8 second time-interval of the MI paradigm. During the training session, we have performed a 5-fold cross validation in order to determine the time window for the best possible classification accuracy by the LDA classifier for classification of the left and right hand MI EEG signals.

In order to explain the working of the EMD method, we have used two single trial EEG signals from the dataset B0103T to obtain IMFs (Fig. 2 and Fig. 3). The left MI EEG signal and its nine IMFs are shown in Fig. 2. Similarly, the Fig. 3 shows the right hand MI EEG signal and its nine IMFs.

The mean frequency was computed for each of the IMFs of the EEG signals corresponding to left and right hand MI tasks. In order to obtain enhanced EEG signals corresponding to left and right hand task, the IMFs whose mean frequencies fall in the range 6-24 Hz were selected. It should be noted that this frequency range covers the μ band (8-13 Hz) and β band (18-24 Hz). These frequency bands are very important for detection of MI EEG signals. The features namely, Hjorth parameters and band powers are then computed for the enhanced EEG signals obtained using the

selected IMFs. In our study we have taken two frequency bands namely 8-12 Hz, and 16-24 Hz for extracting the band powers features and the frequency band 6-24 Hz for the Hjorth features. The extracted features have been given as an input features to the LDA classifier for classification of left and right hand MI EEG signals.

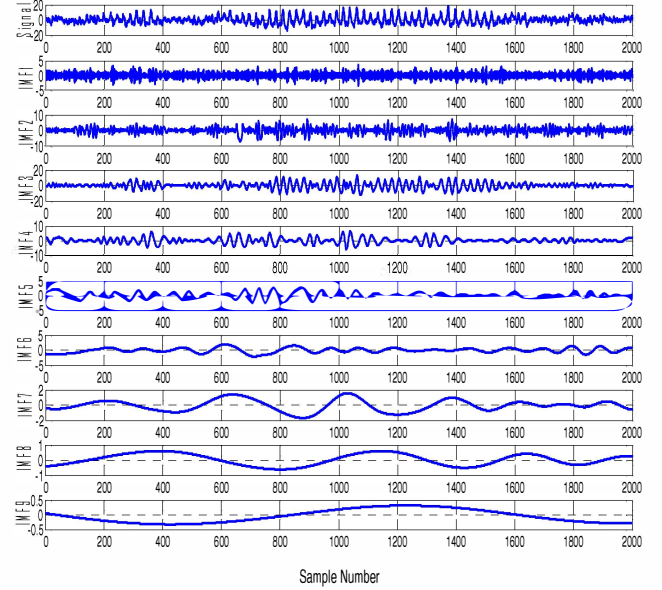


Fig. 2. The EEG signal from C4 of the trial 1 of B0103T with the IMFs generated for the left hand movement.

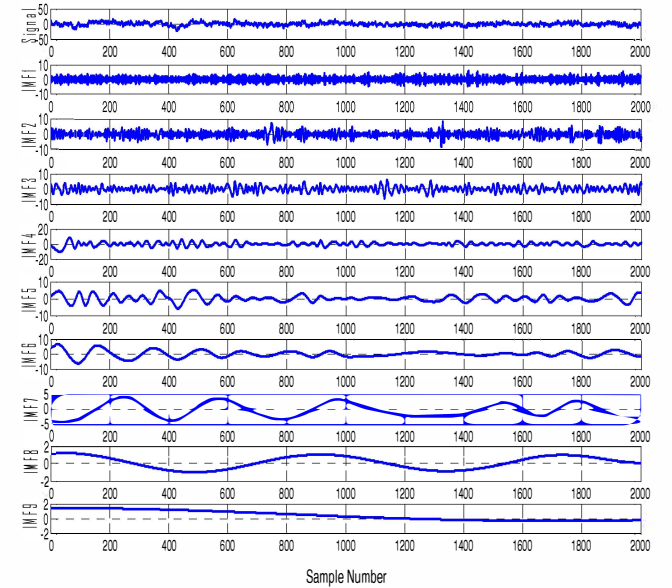


Fig. 3. The EEG signal from C3 of the trial 1 of B0103T with the IMFs generated for the right hand movement.

Table I shows the maximum classification accuracy for BCI competition IV dataset 2B with EMD based filtering, which provides the enhanced EEG signals and with the raw EEG signals, for the nine subjects denoted by B01-B09 across three sessions 03T, 04E, and 05E. It should be noted that only channels C3 and C4 are considered for computing

the results. After applying the EMD based filtering, the group average of the maximum classification accuracy for all subjects across the three sessions improved by 10.54% ($p < 0.001$). In the training session 03T, the results clearly showed the average of the maximum classification accuracy enhanced by 9.65% ($p < 0.05$) when compared the EMD based filtering with the raw EEG signals using the same combination of band powers and Hjorth features.

TABLE I. MAXIMUM CLASSIFICATION ACCURACIES OF THE PROPOSED METHOD BASED ON EMD AND WITHOUT EMD STUDIED ON BCI COMPETITION IV DATASET 2B

Subject	Accuracy with EMD based filtering (%)			Accuracy with raw EEG (%)		
	Training (03T)	Evaluation (04E)	Evaluation (05E)	Training (03T)	Evaluation (04E)	Evaluation (05E)
B01	80	70	55.6	75	60	50
B02	77.5	64.1	70	63.1	58.3	55
B03	98.1	98.1	99.3	52.5	58.7	53.7
B04	90.6	95.6	81.2	95.6	94.3	78.7
B05	85.6	95	97.5	73.1	80.6	77.5
B06	66.2	78.7	71.8	70	77.5	75.6
B07	80.6	68.7	75.6	68.7	61.2	60.6
B08	82.5	85	90.6	82.5	82.5	89.3
B09	85.6	86.2	84.3	79.3	80.6	75.6
Average	82.9	82.4	80.7	73.3	72.6	68.4

In the evaluation sessions, the average improvement with the EMD based filtering is $> 9\%$ ($p < 0.15$) for 04E and in the case 05E, the average classification accuracy improved by $> 12\%$ ($p < 0.01$). Indeed, as shown in Table I, the ten sessions including training and evaluation provided an improvement of $> 10\%$ ($p < 0.001$) classification accuracy with EMD based filtering method across subjects.

Table II shows the average classification accuracy as well as the maximum of the classification accuracy obtained for BCI competition IV dataset 2A for the classification of left and right hand MI EEG signals. We selected C3, C4 and/or Cz channels to compute the classification accuracy. The results in the Table II show the classification accuracies with EMD based filtering and with the raw EEG signals for the nine subjects A01-A09 for both training and evaluation sessions.

TABLE II. MAXIMUM CLASSIFICATION ACCURACIES OF THE PROPOSED METHOD BASED ON EMD AND WITHOUT EMD STUDIED ON BCI COMPETITION IV DATASET 2A

Subject	Accuracy with EMD based filtering (%)		Accuracy with raw EEG (%)		Channel
	Training	Evaluation	Training	Evaluation	
A01	78.6	66.7	52.9	56.3	C3,C4,Cz
A02	68.6	63.9	56.4	54.9	C3
A03	89.3	77.8	62.9	63.2	C3,C4,Cz
A04	72.1	63.2	48.6	55.6	C3,C4
A05	75	72.2	51.4	50	C3
A06	64.3	70.1	57.1	54.2	C3,Cz
A07	78.6	64.6	51.4	50.7	C3,C4
A08	71.4	76.4	56.4	61.1	C3,Cz
A09	77.9	77.1	66.4	74.3	C4
Average	75.1	70.2	57.7	57.8	

Different channel combinations were obtained for each subject based on the optimal accuracies as reported in the Table II. After applying the EMD based filtering, the group average of the maximum classification accuracy across the two sessions was improved by 14.90% ($p < 0.001$). The average of the maximum classification accuracy for training stage showed highly significant improvement of $> 17\%$ ($p < 0.001$). However, in the evaluation session it showed an improvement of $> 12\%$ ($p < 0.001$) in the classification accuracy when compared with the raw EEG signals using the same set of band powers and Hjorth features. The six sessions including training and evaluation showed highly significant improvement of $> 20\%$ ($p < 0.001$), and eight sessions showed improvement in the range of 10% ($p < 0.001$) to 20% ($p < 0.001$) across all the nine subjects. The session-wise p-values for all the nine subjects have been calculated using the repeated measures analysis of variance using *ranova* command in MATLAB.

Fig. 4 and Fig. 5 show the difference between EMD based filtering results and without EMD based filtering. The performance improvement across all the nine subjects are illustrated with the bar graphs. In Fig. 4, the performance improvement for the BCI dataset 2B have been shown for the training and the evaluation sessions respectively. With the proposed EMD based filtering, seven out of nine subjects have shown improvement in the classification accuracy for the training session. There is an improvement in classification accuracy for all the nine subjects using EMD based filtering in the evaluation session 04E. A total of 8 subjects have shown improvement in the classification accuracy using the EMD based filtering in the evaluation session 05E.

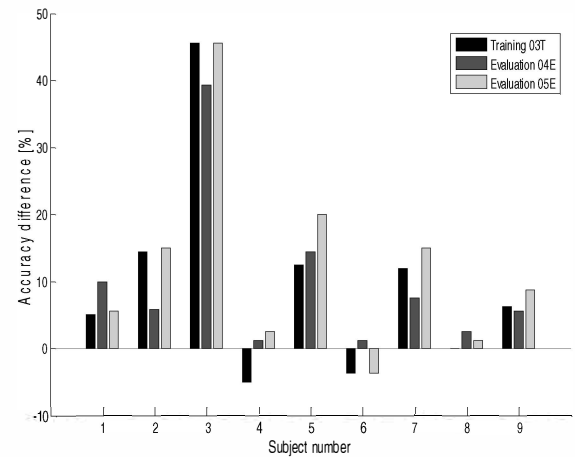


Fig. 4. Accuracy difference between EMD based filtering and raw EEG signals. The bar graphs show an improvement of accuracy when the EMD method is applied in comparison to raw EEG signals across 3 sessions (i.e., training 03T, evaluation 04E and evaluation 05E) for 9 subjects. The data is BCI Competition IV dataset 2B.

In Fig. 5, for the BCI dataset 2A, the classification accuracy have been reported for the training and the evaluation session, respectively. The bar graphs show that the classification accuracies have been significantly improved across the nine subjects using the proposed EMD based filtering.

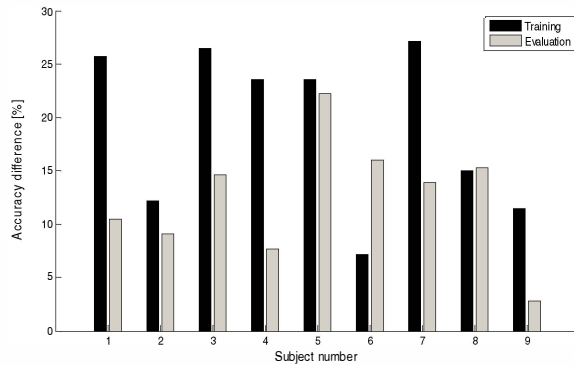


Fig. 5. Accuracy difference between EMD based filtering and raw EEG signals for BCI Competition IV dataset 2A. The bar graphs show an improvement of accuracy when the EMD method is applied in compare to raw EEG signals across 2 sessions (i.e., training T, evaluation E) for 9 subjects.

The bar graphs show that the classification accuracies have been significantly improved across the nine subjects using the proposed EMD based filtering. In both, the training session and the evaluation session, results obtained from all the nine subjects have shown performance improvement $> 7\%$ ($p < 0.001$) in the classification accuracy.

VI. CONCLUSION

We have explored an application of the empirical mode decomposition (EMD) based filtering method for enhancing performance of a motor imagery based brain-computer interface (BCI). A combination of IMFs whose mean frequencies fall in the frequency range of μ and β rhythms has provided improvement in the accuracy of classifying left and right hand MI EEG signals as compared to that without using the EMD method. In future, it would be of interest to develop new features based on the EMD method for classifying MI EEG signals. Although enhanced feature separability offered by the EMD method has helped reduce the performance deterioration due to EEG non-stationarities to some extent, adaptive classification techniques may be investigated for accounting non-stationarities more effectively.

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REFERENCES

- [1] V. Gandhi, G. Prasad, D. Coyle, L. Behera, and T. M. McGinnity, "Quantum neural network-based EEG filtering for a brain – computer interface," *IEEE Trans. Neural Networks Learn. Syst.*, vol. 25, no. 2, pp. 278–288, 2014.
- [2] S. Lemm, B. Blankertz, G. Curio, and K. Müller, "Spatio-spectral filters for improving the classification of single trial EEG," *IEEE Trans. Biomed. Eng.*, vol. 52, no. 9, pp. 1541–1548, 2005.
- [3] H. Zhang, H. Yang, and C. Guan, "Bayesian learning for spatial filtering in an EEG-based brain–computer interface," *IEEE Trans. Neural Networks Learn. Syst.*, vol. 24, no. 7, pp. 1049–1060, 2013.
- [4] M. Arvaneh, C. Guan, K. K. Ang, and C. Quek, "Optimizing spatial filters by minimizing within-class dissimilarities in electroencephalogram-based brain–computer interface," *IEEE Trans. Neural Networks Learn. Syst.*, vol. 24, no. 4, pp. 610–619, 2013.
- [5] D. Coyle, "Neural network based auto association and time-series prediction for biosignal processing in brain-computer interfaces," *IEEE Comput. Intell. Mag.*, vol. 4, pp. 47–59, 2009.
- [6] D. Coyle, G. Prasad, and T. M. McGinnity, "A time-series prediction approach for feature extraction in a brain-computer interface," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 13, no. 4, pp. 461–7, Dec. 2005.
- [7] D. Coyle, G. Prasad, and T. M. McGinnity, "Faster self-organizing fuzzy neural network training and a hyperparameter analysis for a brain – computer interface," *IEEE Trans. Syst. Man, Cybern. B Cybern.*, vol. 39, no. 6, pp. 1458–1471, 2009.
- [8] C. Jia, X. Gao, B. Hong, and S. Gao, "Frequency and phase mixed coding in SSVEP-based brain - Computer interface," *IEEE Trans. Biomed. Eng.*, vol. 58, no. 1, pp. 200–206, 2011.
- [9] N. V. Manyakov, I. Aizenberg, N. Chumerin, and M. M. V. Hulle, "Phase-Coded Brain-Computer Interface Based on MLMVN," in book chapter in *Complex-Valued Neural Networks: Advances and Applications*, 2013, pp. 185–208.
- [10] P. L. Lee, J. J. Sie, Y. J. Liu, C. H. Wu, M. H. Lee, C. H. Shu, P. H. Li, C. W. Sun, and K. K. Shyu, "An SSVEP-actuated brain computer interface using phase-tagged flickering sequences: A cursor system," *Ann. Biomed. Eng.*, vol. 38, no. 7, pp. 2383–2397, 2010.
- [11] N. E. Huang, Z. Shen, S. R. Long, M. C. Wu, H. H. Shih, Q. Zheng, N.-C. Yen, C. C. Tung, and H. H. Liu, "The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis," *Proceedings of the Royal Society of London A: Mathematical, Physical and Engineering Sciences*, vol. 454, pp. 903–995, 1998.
- [12] S. R. H. Davies and C. J. James, "Novel use of empirical mode decomposition in single-trial classification of motor imagery for use in brain-computer interfaces," in *35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, 2013, pp. 5610–3.
- [13] S. R. H. Davies and C. J. James, "Using empirical mode decomposition with spatio-temporal dynamics to classify single-trial motor imagery in BCI," *36th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, pp. 4631–4, Aug. 2014.
- [14] C. Park, D. Looney, N. Rehman, A. Ahrabian, and D. P. Mandic, "Classification of motor imagery BCI using multivariate empirical mode decomposition," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 21, no. 1, pp. 10–22, Jan. 2013.
- [15] C. Park, M. Plank, J. Snider, S. Kim, H. C. Huang, S. Gepshtein, T. P. Coleman, and H. Poizner, "EEG gamma band oscillations differentiate the planning of spatially directed movements of the arm versus eye: multivariate empirical mode decomposition analysis," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 22, no. 5, pp. 1083–96, Sep. 2014.

- [16] J. Saiki, T. Koike, and M. DeBrecht, "Ocular artifacts removal from EEG using EMD," in *Advances in Cognitive Neurodynamics ICCN 2007*. Springer Netherlands, 2008, pp. 527 – 530.
- [17] C. Park, D. Looney, P. Kidmose, M. Ungstrup, and D. P. Mandic, "Time-frequency analysis of EEG asymmetry using bivariate empirical mode decomposition," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 19, no. 4, pp. 366–73, Aug. 2011.
- [18] R. B. Pachori, "Discrimination between ictal and seizure-free EEG signals using empirical mode decomposition," *Res. Lett. Signal Process.*, vol. 2008, pp. 1–5, 2008.
- [19] V. Bajaj and R. B. Pachori, "Classification of seizure and non-seizure EEG signals using empirical mode decomposition," *IEEE Trans. Inf. Technol. Biomed.*, vol. 16, no. 6, pp. 1135–42, Nov. 2012.
- [20] R. Sharma and R. B. Pachori, "Classification of epileptic seizures in EEG signals based on phase space representation of intrinsic mode functions," *Expert Syst. Appl.*, vol. 42, no. 3, pp. 1106–1117, Feb. 2015.
- [21] P. Flandrin, G. Rilling, and P. Goncalves, "Empirical mode decomposition as a filter bank," *IEEE Signal Process. Lett.*, vol. 11, 2004.
- [22] S. Shahid and G. Prasad, "Bispectrum-based feature extraction technique for devising a practical brain-computer interface," *J. Neural Eng.*, vol. 8, no. 2, p. 025014, Apr. 2011.
- [23] H. Raza, G. Prasad, Y. Li, and H. Cecotti, "Covariate shift-adaptation using a transductive learning model for handling non-stationarity in EEG based brain- computer interfaces," in *IEEE International Conference on Bioinformatics and Biomedicine*, 2014, p. 7.
- [24] S. Jose, V. Gandhi, V. Arora, L. Behera, G. Prasad, D. Coyle, and T. M. McGinnity, "EEG denoising with a recurrent quantum neural network for a Brain-Computer Interface," in *International Joint Conference on Neural Networks*, 2011, pp. 1583–1590.
- [25] B. Hjorth, "EEG analysis based on time domain properties," *Electroencephalography and Clinical Neurophysiology*, vol. 29, no. 3, pp. 306–310, 1970.
- [26] S. Shahid, R. Sinha, and G. Prasad, "Mu and beta rhythm modulations in motor imagery related post-stroke EEG: a study under BCI framework for post-stroke rehabilitation," in *BMC Neuroscience*, 2010, vol. 11, no. Suppl 1, p. P127.
- [27] G. Pfurtscheller and F. H. Lopes da Silva, "Event-related EEG/MEG synchronization and desynchronization: basic principles," *Clin. Neurophysiol.*, vol. 110, no. 11, pp. 1842–1857, Nov. 1999.
- [28] F. Lotte, M. Congedo, a Lécuyer, F. Lamarche, and B. Arnaldi, "A review of classification algorithms for EEG-based brain-computer interfaces," *J. Neural Eng.*, vol. 4, no. 2, pp. R1–R13, Jun. 2007.
- [29] A. F. Marquand, S. De Simoni, O. G. O'Daly, S. C. R. Williams, J. Mourão-Miranda, and M. A. Mehta, "Pattern classification of working memory networks reveals differential effects of methylphenidate, atomoxetine, and placebo in healthy volunteers," *Neuropsychopharmacology*, vol. 36, no. 6, pp. 1237–47, May 2011.
- [30] C. Brunner, R. Leeb, and C. Brunner, "BCI Competition 2008 – Graz data set B," pp. 1–6, 2008.
- [31] C. Brunner and R. Leeb, "BCI Competition 2008 – Graz data set A," pp. 1–6, 2008.